

5000 Hydrogen Vehicle Challenge Final Presentation

- **Team/Group Name:** HYDRO DRIVE CODERS (Group 5)
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Understanding the Game

Introduction to HydroDrive 5000, a dynamic game focused on teaching resource management while entertaining users. The game aims to simulate fuel management challenges, integrating learning with fun.



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Key Features of the Game

- Collect hydrogen cells to refuel your vehicle.
- Build stations using earned points to expand gameplay.
- Unlock paths and earn 5000 points before running out of fuel.
- Different difficulty levels to challenge users.
- Movement mechanics with arrow keys.
- Fuel collection and consumption logic



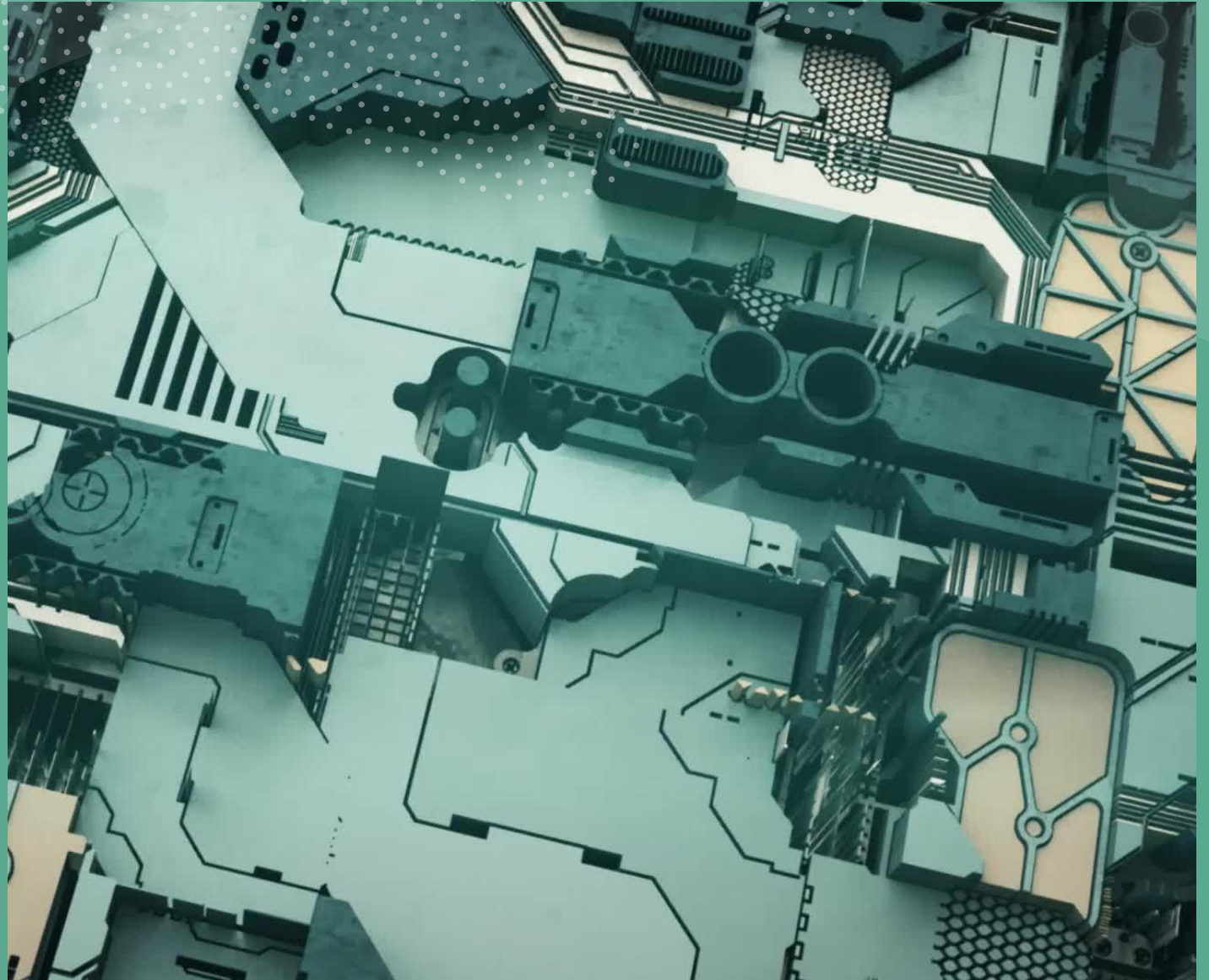
User Experience

- The game offers an engaging user experience with:
- Intuitive controls using arrow keys for movement.
- Real-time feedback through visual and audio cues.
- Vibrant graphics for an immersive gaming environment.



Technical Overview

- Developed in Python using the Pygame library.
- Game logic includes pathfinding, fuel management, and different difficulty scaling.
- Features modular code structure for easy maintenance and upgrades.



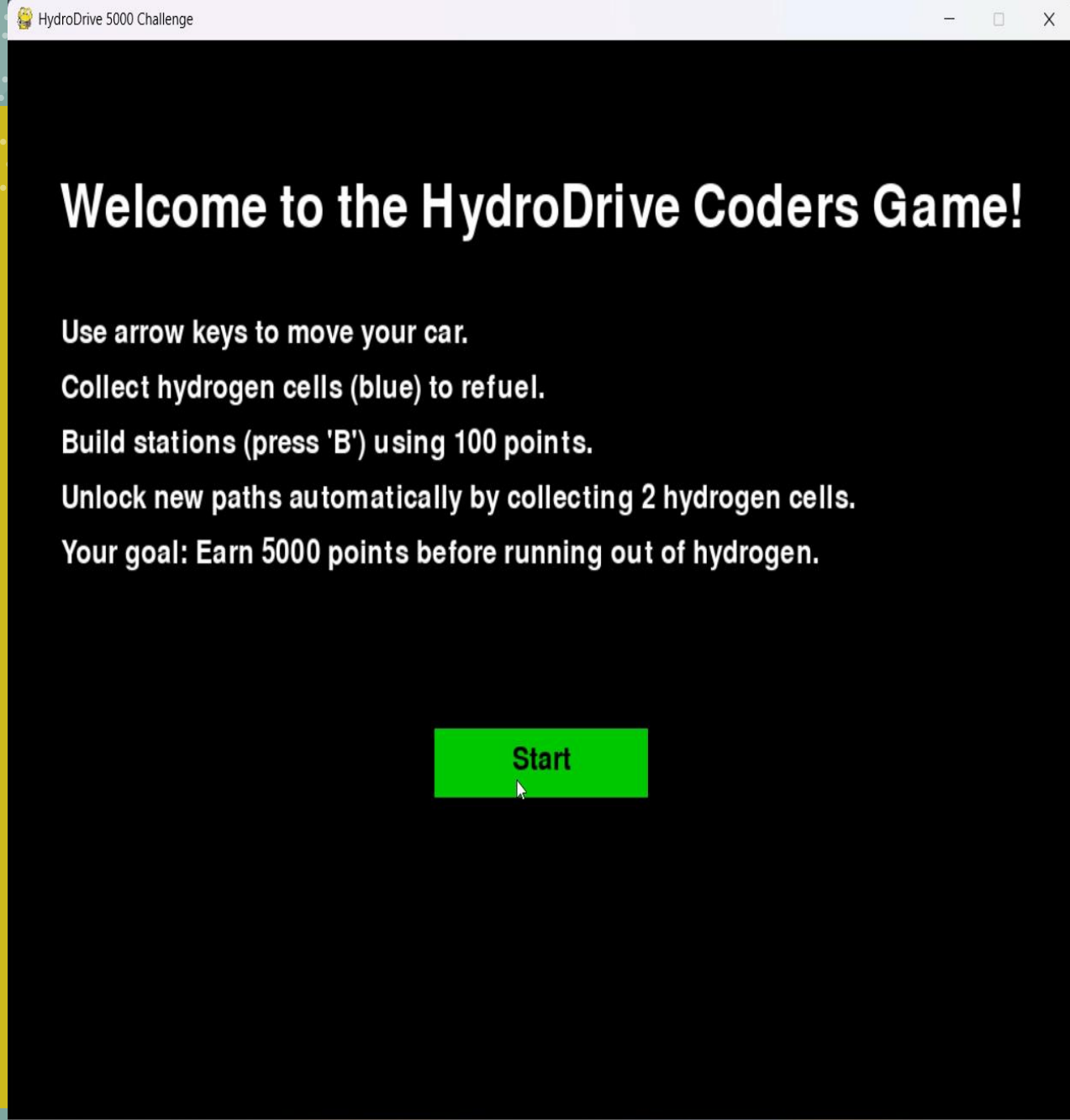
Gameplay Demo

1. Players start with a vehicle and limited hydrogen.
2. Collect hydrogen cells to keep moving.
3. Build stations for additional resources.
4. Unlock walls to access new areas.
5. Earn points to achieve the game's target of 5000 points.



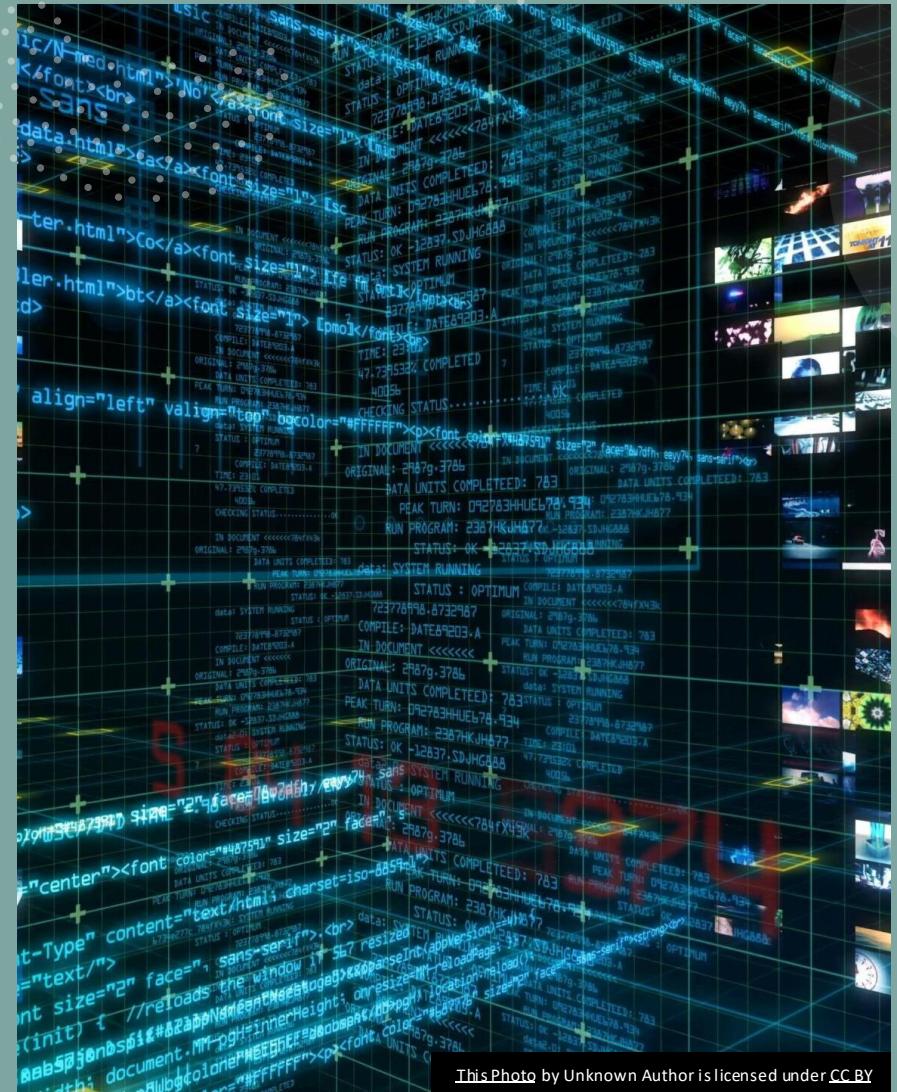
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Game Display Clip



Challenges and Solutions

- Challenge: Managing large datasets for fuel stations.
- Solution: Optimized data structures and caching mechanisms.
- Challenge: Geocoding failures with certain inputs.
- Solution: Implemented error handling to bypass unrecognized entries.
- Challenge: Balancing difficulty levels.
- Solution: Dynamic scaling of hydrogen consumption rates.



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Conclusion

- HydroDrive 5000 successfully combines fun and education, offering players a unique challenge in resource management.

Future improvements could include:

- Multiplayer mode.
- Enhanced graphics and animations.
- Integration with online leaderboards.

